

6 The black pigment melanin, which contributes to hair, skin and eye colour, is produced by cells known as melanocytes.

(a) In people with albinism, the melanocytes do not produce melanin. Albinism is caused by an inherited gene mutation.

(i) Outline how a gene mutation may occur.

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(ii) Albinism is an autosomal recessive condition.

Explain what is meant by the term *recessive*.

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(iii) Using appropriate symbols, draw a genetic diagram to show how a man and a woman, who both produce melanin, could have a child with albinism.

(b) Melanin is produced by the action of the enzyme tyrosinase on the amino acid tyrosine.

A study was carried out to investigate the effect of an extract of the starfish *Patiria pectinifera* on the activity of tyrosinase.

Table 6.1 shows the results of this study.

Table 6.1

concentration of starfish extract / $\mu\text{g cm}^{-3}$	percentage tyrosinase activity
0	100
4	90
8	77
16	68
32	56
64	46
128	32

Suggest how the starfish extract affects the activity of tyrosinase.

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